Chihye Han (Kelsey)

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Education

Johns Hopkins University — Baltimore, MD
Ph.D. in Cognitive Science, Computational Track — 2020–Present
M.A. in Cognitive Science — 2022
Advisor: Michael F. Bonner

Korea Advanced Institute of Science and Technology — Daejeon, Korea
M.S. in Electrical Engineering — 2017–2019
Advisor: Dae-Shik Kim

Carleton College — Northfield, MN
B.A. in Cognitive Science, Neuroscience Concentration — 2009–2013
Cum laude, Honors in Music Performance
Advisors: Kathleen G. Galotti and Roy Eleveton

Publications

Han, C. & Bonner, M.F. (2026). High-dimensional structure underlying individual differences in naturalistic visual experience. *Current Biology*.

Park, G., **Han, C.**, Yoon, W., & Kim, D. (2020). MHSAN: Multi-Head Self-Attention Network for Visual Semantic Embedding. *2020 IEEE Winter Conference on Applications of Computer Vision* (WACV).

Han, C., Yoon, W., Kwon G., Nam, S., & Kim, D. (2019). Representation of White- and Black-Box Adversarial Examples in Deep Neural Networks and Humans: A Functional Magnetic Resonance Imaging Study. *2019 International Joint Conference on Neural Networks* (IJCNN).

Kwon, G., **Han, C.**, & Kim, D. (2019). Generation of 3D Brain MRI Using Auto-Encoding Generative Adversarial Networks. *2019 Medical Image Computing and Computer Assisted Intervention* (MICCAI).

Hong, J., Li, L., **Han, C.**, Jin, B., Yang, Q., & Yang, Z. (2016). Optimizing Hadoop Framework for Solid State Drives. *2016 IEEE International Congress on Big Data* (BigData Congress).

Conference Presentations

Posters

Han, C., & Bonner, M. F. Low-variance dimensions of cortical activity carry behaviorally relevant information. *Vision Sciences Society*; May 15–19, 2026; St. Petersburg, FL.

Han, C., Gauthaman, R.M. & Bonner, M. F. Behavioral relevance of high-dimensional neural representations. *Cognitive Computational Neuroscience*; Aug 12–15, 2025; Amsterdam, The Netherlands.

Han, C. & Bonner, M. F. High-dimensional structure underlying individual differences in naturalistic visual experience. *Vision Sciences Society*; May 16–20, 2025; St. Petersburg, FL.

Han, C. & Bonner, M. F. High-dimensional latent manifolds and individual differences in naturalistic movie viewing. *Cognitive Computational Neuroscience*; Aug 12–15, 2024; Boston, MA.

Han, C. & Bonner, M. F. High-dimensional latent manifolds as predictors of individual differences in naturalistic movie viewing. *Vision Sciences Society*; May 17–22, 2024; St. Petersburg, FL.

Han, C., Magri, C., & Bonner, M. F. Quantifying the latent semantic content of visual representations. *Vision Sciences Society*; May 21–26, 2021; Virtual.

Han, C., Yoon, W., Nam, S., & Kim, D. Neural Representation of Adversarial Images: An fMRI Study. *Women in Machine Learning Workshop*; Dec 3, 2018; Montreal, Canada.

Park, J., **Han, C.**, Kim, M., & Kim, D. End-to-End rs-fMRI Data Classification Using Deep Convolutional and Long Short-Term Memory Networks. *Organization for Human Brain Mapping*; Jun 17–21, 2018; Singapore.

Kim, M., **Han, C.**, Park, J., & Kim, D. T1 Image Synthesis with Deep Convolutional Generative Adversarial Networks. *Organization for Human Brain Mapping*; Jun 17–21, 2018; Singapore.

Talks

Han, C., Yoon, W., Kwon G., Nam, S., & Kim, D. Representation of White- and Black-Box Adversarial Examples in Deep Neural Networks and Humans: A Functional Magnetic Resonance Imaging Study. *International Joint Conference on Neural Networks*; Jul 14–19, 2019; Budapest, Hungary.

Invited Talks

Cold Spring Harbor Laboratory, Klindt Laboratory — June 2026
Host: David Klindt

Dartmouth College, Functional Imaging and Naturalistic Neuroscience Lab — May 2026
Host: Emily S. Finn

National Institutes of Health, Unit on Learning and Plasticity, Laboratory of Brain and Cognition — May 2026
Host: Chris I. Baker

Institute for Basic Science, Perceptual and Cognitive Neuroscience Lab — Apr 2026
Host: Won Mok Shim

Freie Universität Berlin, Neural Dynamics of Visual Cognition Group — Apr 2026
Host: Radoslaw M. Cichy

Johns Hopkins University, Chen & Honey Lab — Apr 2026
Host: Janice Chen & Christopher J. Honey

Johns Hopkins University, OneNeuro Student Talk — Oct 2025

Honors & Awards

Elsevier/*Vision Research* Travel Award (*V*-VSS) — 2021
National Scholarship (KAIST) — 2017–2019
Student Travel Award (International Joint Conference of Neural Networks) — 2019
Student Travel Award (Women in Machine Learning) — 2018
Best Paper Award (International Congress on Big Data) — 2016
Value Creator Award (Samsung Human Resources Development Center) — 2014
Sigma Xi Nomination (Carleton College) — 2012
Robert J. Kolenkow and Robert A. Reitz Fund for Undergraduate Research — 2010

Teaching

Teaching Assistant, Johns Hopkins University
Computational Cognitive Neuroscience of Vision — Spring 2024
Cognitive Neuropsychology — Fall 2021, Fall 2023
Cognitive Neuropsychology in Vision — Spring 2022
Cognitive Neuroscience — Spring 2021
Teaching Assistant, KAIST
Electronics Design Lab — Spring 2019
Neural Networks — Fall 2018
Teaching Assistant, Carleton College

Music Theory I & II Fall–Winter 2012

Professional Experience

LG AI Research Seoul, Korea
AI R&D Strategist 2022–2023

KAIST Daejeon, Korea
Research Intern, hosted by Sang Ah Lee 2020

OBELAB Seoul, Korea
Analysis Engineer 2017

Samsung Electronics Seoul, Korea
Software Engineer 2014–2016